

Atharv Sandeep Patil

Los Angeles, CA | +1 (213) 498-9020 | atharvsa@usc.edu | LinkedIn | Github | Portfolio

EDUCATION

Master of Science in Computer Science, University of Southern California

Aug 2026 – May 2028 | Los Angeles, CA

Bachelor of Technology (Computer Science & Engineering)

Aug 2020 – Jan 2024 | Surat, India

Sardar Vallabhbhai National Institute of Technology

CGPA: 8.75/10 (First Class with Distinction)

TECHNICAL SKILLS

Foundations: Data Structures & Algorithms, Object-Oriented Design, Distributed Systems, Performance Optimization

Programming: Java, Python, JavaScript, SQL, Shell/Bash

Backend & Frontend: Spring Boot, NodeJS, Microservices, Multithreading, ReactJS

Cloud & DevOps: AWS (Lambda, S3, ECS, OpenSearch), Docker, Google ADK, Spinnaker, CI/CD, Unit Testing, Code Review, Git

Databases: Oracle DB, MongoDB, Neo4j

AI Engineering: LLM, RAG Pipelines, Prompt Engineering, Agent Development, Machine Learning

PROFESSIONAL EXPERIENCE

Software Development Engineer II, J.P. Morgan Chase & Co.

Jan 2026 – Jun 2026 | Mumbai, India

- Partnered with stakeholders to extend a RAG system across 4 specialized sub-agents with a client-in-context feature, saving advisors ~3 minutes of manual UI navigation to retrieve client-specific data.
- Led LLM model migration from GPT 4o to GPT 5.2, improving agent response accuracy by 5% (90% to 95%) via prompt re-engineering and automated validation.
- Spearheaded migration from ChromaDB to AWS OpenSearch, improving retrieval architecture scalability and maintainability.

Software Development Engineer, J.P. Morgan Chase & Co.

Jan 2024 – Jan 2026 | Mumbai, India

- Re-architected a high-performance offset calculator (Spring Boot + AWS Lambda + multithreading) processing ~600K records, cutting runtime from ~2 hrs to ~8 min (96% faster) and automating report delivery via a React dashboard.
- Diagnosed and resolved LLM hallucination and retrieval issues in the RAG system serving ~200 advisors, improving reliability.
- Moved ~15 years of legacy mainframe data into a distributed data mesh, working within Lambda's memory and timeout limits.
- Migrated the RAG service from AWS EKS to AWS ECS, reducing infrastructure costs by ~30%.
- Engineered Spring Boot microservices for investment account management, enabling new product launches for ~500K clients.

Software Engineer Program Intern, J.P. Morgan Chase & Co.

May 2023 – Jul 2023 | Mumbai, India

- Optimized a parser translating internal query language into SQL using Spring Boot, extending support for Databricks SQL dialect and eliminating compatibility errors.
- Constructed a nested data retrieval method in Databricks using Apache Calcite, reducing database round trips from 3 to 1.

RESEARCH EXPERIENCE

Undergraduate Researcher , Sardar Vallabhbhai National Institute of Technology

Jul 2023 – Dec 2023 | Surat, India

- Designed and fine-tuned an intent detection module using Wit.ai to classify legal queries, enabling accurate Cypher query generation and information retrieval of 9,377 Supreme Court cases from a Neo4j-based legal property graph.
- Evaluated an NLU-based approach against BERT and BiLSTM baselines, achieving a significant improvement in intent detection (F1 Score: 0.89) while maintaining competitive entity extraction accuracy.
- Publication: Procedia Computer Science, presented at ICMLDE 2025.

PROJECTS

EvalGate , Personal Project

- Built an evaluation and regression-testing framework for LLM agents supporting 5 scorer types (contains, regex, numeric, trajectory, LLM-judge), running N-trial statistical scoring with Wilson confidence intervals to handle output variance.
- Designed a two-condition regression gate (pass-rate margin AND confidence-interval exclusion) and validated against hand-computed boundary cases, including a 6/8 to 5/8 pass-rate drop correctly identified as statistical noise rather than a real regression.
- Deployed a self-built multi-agent research assistant (1 coordinator, 3 sub-agents) on Google ADK, integrating the live SEC EDGAR API and shipping a GitHub Actions CI gate to block regressive pull requests.

Healthcare Platform , Code For Good - JPMC Hackathon

- Delivered a full-stack platform used by ~200 healthcare workers to remotely track and counsel mothers in rural India, with smart logic to aid workers in selecting suitable medicines and vaccines.
- Collaborated on backend development employing NodeJS, Express, MongoDB, and Firebase authentication, ensuring smooth functionality and data protection.

CERTIFICATIONS & LEADERSHIP

- Amazon AWS Certified Developer – Associate (Apr 2026)
- Kaggle 5-Day AI Agents Intensive Coding Course (Jun 2026)
- Web Development Lead - Google Developer Student Clubs, NIT Surat (Aug 2022- Apr 2023)
- Logistics Lead - Hack The Tank, 24-hour Pan-India Hackathon ft. Shark Tank India entrepreneurs with 100+ participants (Mar 2023)
- Video Editor Lead - Renesa, Media and Publication House, NIT Surat (Aug 2022- Apr 2023)